Ziwen Hua

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Education

09/2026 - Present **University of Maryland, College Park, MD, USA**
Undergraduate Student, Major Undeclared

09/2022 - 06/2025 **Shanghai No.2 High School, Meilong Campus, Shanghai, China**
Grade 10 - Grade 12, High School Diploma

Internship

Shanghai HintonAI Tech 07/2026 - 08/2026

- Designed and improved the hand-eye calibration module for a ROS2-based open source robotic painting system, RobRoss.
- Collaborated with the path-execution pipeline to ensure calibration outputs were robust in real robot runs, reducing coordinate drift and alignment errors during autonomous drawing.

Research & Projects

Research on Preparation of Rapid Cooling Equipment 09/2023 - 06/2024
Shanghai Jiaotong University

- Read papers, learned college physics and industrial cooling technology, and studied methods for physics research.
- Produced a mini semiconductor refrigerator in the lab and wrote a paper.

Application of Generative Pre-trained Transformer Models 09/2023 - 06/2024

- Researched the use of modern AI models in daily life, learned Python programming and knowledge of advanced mathematics (i.e. Bayesian Rule).
- Trained a small AI model to translate Chinese into English.

Survey on Community Care on Shanghai Metro 09/2022 - 06/2023

- Interviewed passengers, conducted questionnaire survey, analyzed questionnaires, and prepared survey reports.
- Proposed improvements for human-centered urban rail development.

Extracurricular Activities

Student Union Discipline Committee, Officer 09/2022 - 06/2024

- Oversaw the discipline of c.800 students in the evening self-study session to ensure minimal distraction; prepared a weekly discipline assessment report.

School TV Station, Photographer & Video Recorder 09/2022 - 06/2024

- Worked as the only drone operator, recorded major events for the school's official WeChat website and publicity brochures.

Agricultural Technology Learning, Member 11/2023

- Completed a plant biodiversity survey and learned how to protect plants, such as by controlling pests and diseases, as well as fertilization and weeding techniques and how to cook local dishes.

Volunteer Activities

Shanghai Aerospace Enthusiasts Center, Volunteer 06/2023 - 08/2023

- Assisted staff in maintaining order during weekends and summer vacations; Served as a tour guide and TA in activities like flying simulation and making plane models; Explained theory of various machines to primary school students visited the center, to them.

Shanghai No.2 High School-Campus Open Day, Volunteer 06/2023

- Introduced school's history, campus life, club activities and useful tips; Talked about what school's STEM program involved; Showed my program work - a robotic car using autonomous driving algorithms to move.

Honors & Awards

Third Prize, 'China Middle School Cup' Youth AI Competition 12/2023
Second Prize, Wuben Scholarship, Shanghai No.2 High School 06/2024
Outstanding Student Union Cadre, Shanghai No.2 High School 12/2023
Third Prize, Drone Obstacle Flight Racing, School Science Festival 05/2024

Skills & Hobbies

- Self-study "Dive into Deep Learning", a textbook of UCB(Learned basic algorithms of modern AI and the procedure to train an AI model)
- **Programming Skill:** Python, C++
- **Hobbies:** Swimming, Photography